

Preamble Guide

This document explains the structure of the LaTeX preamble used as the base template. Each section below corresponds to a block in the preamble and includes what it does, why it is useful, and the commands that depend on it.

1. Description

This file documents the preamble that is used as a shared base for multiple LaTeX documents. Other files include it with `\input{preamble}`, so changing the preamble modifies every document that imports it. The preamble includes packages, page settings, colors, frames, hyperlinks, and custom commands used across guides.

2. General

- Document type: `standalone` is used with `varwidth` to produce compact pages and allow custom margins.
- Margins: `geometry` sets custom margins for the page layout.
- Line spacing: `\linespread{1.2}` increases readability.
- Input encoding: `inputenc` with `utf8` supports accented characters.

```
\documentclass[varwidth, margin=2cm, 12pt]{standalone}
\usepackage[left=2cm, right=-8cm, top=1cm, bottom=2.5cm]{geometry}
\linespread{1.2}
\usepackage[utf8]{inputenc}
```

3. Tables

- `multicol`: allows text or content to be split into columns.
- `longtable`: supports tables that span multiple pages.
- `array`: provides extra control over column formatting.
- `booktabs`: improves table rules and spacing for cleaner design.

```
\usepackage{multicol}
\usepackage{longtable}
\usepackage{array}
\usepackage{booktabs}
```

4. Mathematics

- `amsmath`: improves equation environments and alignment.
- `amssymb` and `amsfonts`: provide additional math symbols and fonts.
- `mathtools`: extends `amsmath` with extra tools and fixes.

```
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsfonts}
\usepackage{mathtools}
```

5. Hidden Content Switch

This section defines a boolean switch to show or hide explanatory content without deleting it.

- `\newboolean{explicaciones}` creates the switch.
- `\mostrarExplicaciones{true/false}` turns explanations on or off.
- `\explicacion{A}{B}` prints A when the switch is true, and B when it is false.

```
\newboolean{explicaciones}
\newcommand{\explicacion}[2]{\ifthenelse{\boolean{explicaciones}}{#1}{#2}}
\newcommand{\mostrarExplicaciones}[1]{\setboolean{explicaciones}{#1}}
```

Example usage:
This text will appear.

6. Numbering and Table of Contents

- `fancyhdr` is loaded, but page numbering is disabled with `\pagestyle{empty}`.
- `tocloft` customizes the table of contents.
- Dots and page numbers are removed for sections and subsections.

```
\usepackage{fancyhdr}
\pagestyle{empty}

\usepackage{tocloft}
\renewcommand{\cftdot}{\cftdotfill{\cftdotsep}}
\cftpagenumbersoff{section}
\cftpagenumbersoff{subsection}
```

7. Images, Figures, and Colors

7.1. Images and positioning

- `graphicx` enables `\includegraphics`.
- `float` allows better control of figure placement.
- `subcaption` enables subfigures.
- `\graphicspath{{imgs}}` sets the default image folder.

```
\usepackage{graphicx}
\usepackage{float}
\graphicspath{{imgs}}
\usepackage{subcaption}
```

7.2. TikZ figures

- `tikz` supports vector graphics drawn in LaTeX.
- `arrows.meta` adds arrow styles and tools.

```
\usepackage{tikz}
\usetikzlibrary{arrows.meta}
```

7.3. Color palette

This palette defines named colors used across frames, background, and hyperlink styling.

```
\usepackage{xcolor}
\definecolor{popUp}{HTML}{666666}
\definecolor{backgroundC}{HTML}{D0E8F2}
\definecolor{backgroundCC}{HTML}{FFFFFF}
\definecolor{links}{HTML}{CC6F5F}
```

8. Frames

Frames are used to highlight content blocks and code blocks in a consistent style.

- `tcolorbox` provides advanced box tools and is used for `\fboxC`.
- `adjustbox` and the `\fboxrule` / `\fboxsep` lengths tune borders and padding.
- `mdframed` defines two main styles:
 - `estiloGeneral`: main content frame with rounded corners.
 - `code`: code frame with a light gray background.

```
\mdfdefinestyle{estiloGeneral}{...}
\mdfdefinestyle{code}{...}
\newtcbbox{\fboxC}{...}
```

This is a sample frame using `estiloGeneral`.

This is a sample frame using `code`.

9. Page Settings

9.1. Background commands

- `\background` paints the page background using TikZ and `backgroundC`.
- `\backgroundC` paints an alternative background using `backgroundCC`.

```
\newcommand{\background}{...}
\newcommand{\backgroundC}{...}
```

9.2. Page width macro

- `\anchoPag` defines the minipage width used in documents.

```
\newcommand{\anchoPag}[0]{20cm}
```

10. Font and Code Listings

- `tgbonum` sets the main text font.
- `listings` is used for code formatting, including a custom Python style.
- `babel` is configured for Spanish in this preamble, affecting hyphenation and language rules.

```
\usepackage{tgbonum}
\usepackage{listings}
\usepackage[spanish]{babel}
\selectlanguage{spanish}
\lstdefinestyle{python}{...}
```

11. Hyperlinks and Custom Link Commands

11.1. Hyperref configuration

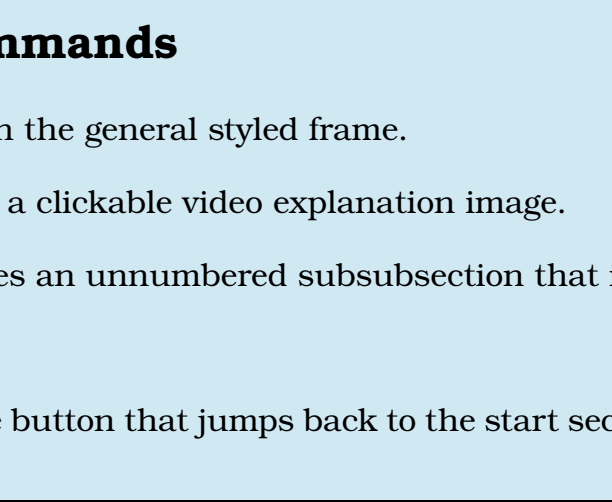
- `hyperref` enables clickable links.
- `\hypersetup` defines link colors using the color palette.

```
\usepackage{hyperref}
\hypersetup{
  colorlinks=true,
  linkcolor=links,
  filecolor=magenta,
  urlcolor=brown,
}
```

11.2. Custom link commands

- `\bigLink{url}{text}` creates a centered framed large hyperlink.
- `\smallLink{url}{text}` creates a centered framed normal hyperlink.
- `\bfLink{url}{text}` creates a bold clickable text link.
- `\smallUrl{url}` prints a framed URL centered on the page.

Examples:



12. Custom Figure Commands

These commands wrap `\includegraphics` with consistent spacing and optional framing, and can optionally create clickable images.

- `\espacioImagenes` controls horizontal offset applied to figures.
- `\fig` inserts an image without a frame.
- `\ffig` inserts an image in a figure environment with a visible frame.
- `\hfig` inserts a framed image that links to a URL.
- `\hffig` inserts an unframed image that links to a URL.

```
\newcommand{\fig}[3][\espacioImagenes]{...}
\newcommand{\ffig}[2]{...}
\newcommand{\hfig}[3]{...}
\newcommand{\hffig}[3]{...}
```

13. Other Utility Commands

- `\cuadro{...}` wraps content in the general styled frame.
- `\linkExplicacion{url}` places a clickable video explanation image.
- `\subSecLink{url}{text}` creates an unnumbered subsubsection that is a hyperlink.
- `\esp` adds vertical spacing.
- `\secInicio` creates a clickable button that jumps back to the start section labeled `sec:inicio`.

Example:
This content is inside a `\cuadro` frame.

mdframed package

14. Navigation and Visual Examples

- The command `\secInicio` depends on a label called `sec:inicio`.
- If your document includes a table of contents section labeled `sec:inicio`, the button will jump there.

Below is a visual example of a clickable image link:

